AI Study Companion

This plan is AI-generated. Make sure to review and change it to ensure accuracy. Learn more.

Business problem

This project is a multifaceted AI-powered study application designed to enhance student learning by transforming physical or digital notes into structured, personalized study materials. The app allows users to scan their study pages using their mobile device, after which the system analyzes the content using natural language processing and AI integration (such as ChatGPT) to identify the topic and suggest relevant courses within that subject area. Based on user selection, the app

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Purpose of this plan

This plan serves as a blueprint for addressing the business problem described above, outlining the user requirements, data model, and technology that will be used to get things done.

User requirements

Student

A learner who uses the app to scan notes, generate study materials, access tutoring, and plan courses.

As a user, I need to:

- Scan physical or digital notes so that I can convert them into structured study materials.
- Generate regular study sheets in formats like Cornell notes so that I can review content in an organized way.
- Produce cram sheets for each topic so that I can quickly review key concepts before exams.
- Create flashcards from my notes so that I can use spaced repetition techniques.
- Store all generated materials in class-specific folders so that I can easily retrieve and organize content.

- Receive AI-powered recommendations for relevant courses based on my scanned notes so that I can align my study materials with course objectives.
- Access adaptive study suggestions based on course difficulty and my progress so that I can study more efficiently.
- Interact with an AI tutor chatbot for real-time academic assistance so that I can get help whenever I need it.
- Plan my semester using integration with course catalogs and assignment breakdowns so that I stay on track academically.
- Connect to campus resources like tutoring and writing centers so that I can access additional support.
- Provide feedback on my understanding of topics so that the system can tailor future recommendations.

Tutor

Person who provides academic support, offers guidance, and interacts with students via the app.

As a user, I need to:

- Interact with students via the app so that I can provide real-time guidance.
- Review student progress and recommend improvements so that students benefit from targeted academic support.

Learner

Learner who uses the mobile app to digitize, analyze, and organize study materials.

As a user, I need to:

- Digitize physical or digital notes so that I can organize and analyze my study materials.
- Access AI-analyzed summaries of my notes so that I can focus on key topics.
- Organize study materials into personalized folders so that I can easily retrieve them when needed.

Developer

Person responsible for building and maintaining the app's functionality.

As a user, I need to:

- Build new app features based on user feedback so that the app evolves to meet student needs.
- Maintain app stability and performance so that users have a seamless experience.
- Implement integration with AI and course data so that personalized study materials are generated.

AI data specialist

Person managing AI integration, NLP analysis, and content formatting.

As a user, I need to:

- Integrate and manage AI powered content analysis so that note processing is accurate and efficient.
- Develop NLP models for topic identification so that students receive relevant course recommendations.
- Format generated materials to match study output templates so that content is easy to use.

Designer

Person responsible for user interface, experience, and visual consistency.

As a user, I need to:

- Design user interface layouts so that the app is intuitive for students.
- Ensure visual consistency across all screens so that the user experience is seamless.
- Collaborate with developers and AI specialists so that designed features fit technical constraints.

Presenter

Person who uses the app to generate materials for academic presentations and organize content to share with others.

As a user, I need to:

- Generate presentation materials from notes so that I can convey study concepts to others.
- Organize and access content for academic presentations so that sharing information is streamlined.

User processes

The processes for solving the business problem. Each process appears as a card next to this document.

Digitize and Analyze Study Materials

This process covers the scanning or uploading of physical and digital notes, AI-powered content analysis, topic identification, and the generation of personalized study materials. It includes organizing outputs into folders and recommending relevant courses based on analyzed content.

Adaptive Study Support and Academic Planning

This process focuses on providing adaptive study recommendations, AI tutor chatbot assistance, academic planning using course catalogs and assignment breakdowns, and connecting students to campus resources. It also includes feedback collection to tailor future recommendations.

Presentation and Content Sharing

This process enables users to generate, organize, and access materials for academic presentations, facilitating the sharing of study concepts and content with others.

App Design, Development, and Maintenance

This process encompasses the design, development, and ongoing maintenance of the app, ensuring seamless integration of AI, user interface consistency, and responsiveness to user feedback.

Data model

The tables that will be used to store and provide information.

User

This table contains records of users including students and tutors.

Study Material

This table contains digitized study materials scanned by users.

Topic

This table contains main topics identified from study materials.

Course

This table contains course information related to study topics.

Study Sheet

This table contains study sheets created from study materials.

Folder

This table contains folders to organize study sheets and flashcards.

Flashcard

This table contains flashcards created from study materials.

Flashcard Difficulty

This table categorizes flashcards by difficulty level.

Tutoring Service

This table contains campus tutoring and writing center offerings.

Feedback

This table contains user feedback on topics and materials.

Technology

The apps, flows, and other objects that will be used to solve the business problem.

Smart Study Mobile
Canvas app New
Students and learners scan notes, generate study sheets, cram sheets, flashcards, and organize materials in class folders for efficient, personalized studying.

Academic Advisor Portal
Model-driven app New
A portal for tutors to provide live guidance, review student progress, assign resources, and recommend study improvements for targeted support.

Course Progress Insights
Report New
An interactive report for students and presenters to track course alignment, study habits, and feedback trends, enhancing academic planning.

Study Material Exporter
Power Automate flow New
Automates export of flashcards and study sheets to Quizlet or email for easier review and collaborative studying for all users.

AI Study Assistant
Agent New
An intelligent chatbot that analyzes scanned materials, identifies topics, suggests courses, and assists students with academic queries in real time.

PersonalizationFeedback Bot
Agent New

Gathers student feedback on topic understanding and tailors recommendations for more effective learning pathways.

AI Content Curation
Agent New

Automates note analysis, NLP topic discovery, content formatting, and template generation for developers, designers, and data specialists.